

ACTIVITY 13 — LEARNER WORKSHEET**Cost Estimating, Budget and Reserves**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	90 minutes	LO2	3 - Plan the Project

What you are doing

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

What you will produce: cost estimates, cost baseline, reserve analysis — submitted to artifacts/.

What you need before you start

- Lab 11 WBS work packages and effort estimates
- Lab 12 schedule
- Lab 05 business case
- Lab 06 budget summary

Your steps

1. Build the bottom-up estimate from the WBS. Create artifacts/13-cost-estimates.md. Take effort from the Lab 11 WBS dictionary and apply the differentiated resource rates.
2. Contingency reserve versus management reserve. This is the central concept of the lab and one of the most reliably tested distinctions in the entire cost area. Learn it as a table and be able to reproduce it.
3. Size the contingency reserve by expected monetary value. A reserve set as "10% because that is what we always do" is not an estimate; it is a habit. Size it from the risk register.
4. Build the time-phased cost baseline. A total budget is not a baseline. A baseline is time-phased - it says how much should have been spent by any given date, which is what makes variance measurable. Plotted cumulatively it produces the characteristic S-curve: slow at the start while the team ramps and designs, steep through the build, flattening at the end as work completes.
5. Funding limit reconciliation and cash flow. Contoso's board releases funds quarterly:
6. Analyse the cost of quality
7. Answer the exam-style scenarios

Your working space

Complete every field. Keep this worksheet — it is your reference for the rest of the course.

Step 1 — Build the bottom-up estimate from the WBS

Step 2 — Contingency reserve versus management reserve

Step 3 — Size the contingency reserve by expected monetary value

Step 4 — Build the time-phased cost baseline

Step 5 — Funding limit reconciliation and cash flow

Step 6 — Analyse the cost of quality

Step 7 — Answer the exam-style scenarios

Deliverable checklist

- 13-cost-estimates.md - the bottom-up estimate from the Lab 11 work packages with role, effort, rate and cost per package; the reconciliation to the charter's SGD 348,000 development line with the variance stated; the non-labour categories with their estimating basis named.
- The ROM and definitive ranges applied to SGD 480,000, with the finding that the definitive upper bound exceeds the ceiling.
- 13-reserve-analysis.md - the contingency versus management reserve table; the EMV computation across at least 16 risks showing gross and residual exposure; the reconciliation to SGD 26,000 with the variance stated; the analysis of the no-management-reserve condition as a risk in itself.
- 13-cost-baseline.md - the 12-period time-phased baseline with planned spend and cumulative PV per period, summing to SGD 480,000, with PV at week 14 equal to SGD 268,000; the S-curve plotted; the convention for contingency stated explicitly.
- The funding limit reconciliation with the Q2 shortfall identified and an action recorded.
- The cost of quality analysis across all four categories with real figures, including the accessibility prevention-versus-retrofit ratio.
- Written answers to the three scenarios.

